XYZ MUSIC: PREDICTING PREMIUM USER CONVERSION

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The project aims to use predictive model to precisely identify XYZ's potential premium users and boost subscription conversions.

Project Introduction

XYZ Company Overview

XYZ Music is a social music platform operating under a freemium model. The website offers basic services for free, while premium users enjoy advanced features with a monthly subscription.

Project Background

However, based on the previous marketing campaign data, only about 4% of free users upgraded to premium, leading to inefficient marketing spend.

Today, we are delivery and report on ...

Predictive Conversion Model

- Our best model predicts which users are most likely to upgrade to premium.
- Helps XYZ target potential subscribers more efficiently.

User Insight

- Understanding key behavioral and social features that drive premium adoption.

Marketing Impact & ROI

- How the model helps reduce campaign cost and improve return on investment.

Transforming raw campaign data into a reliable training dataset

Dataset

- 41,540 users from previous campaign
- 25 attributes capturing user activity, engagement, and demographics
- Only 4% converted → campaign data is highly imbalanced

Data Preparation

- Removed irrelevant fields and missing data
- Balanced training data to ensure fair learning between adopters and non-adopters

Modeling Approach

- The refined dataset was then used to test multiple machine learning models
  - KNN
  - Decision Tree
  - Random Forest
  - Random Forest(Ranger)

Best Model Result: Smarter Targeting with Ranger

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Modeling Approach

- Random Forest (Ranger) BEST
  - most stable and reliable performance among all candidates.

Best Model Result: Smarter Targeting with Random Forest(Ranger)

Why F1 and AUC?

- F1: A good indicator in not SMOTEd validation data set
- AUC: Stable even when class proportion are skewed

Ranger (Fast, optimized Random Forest) →

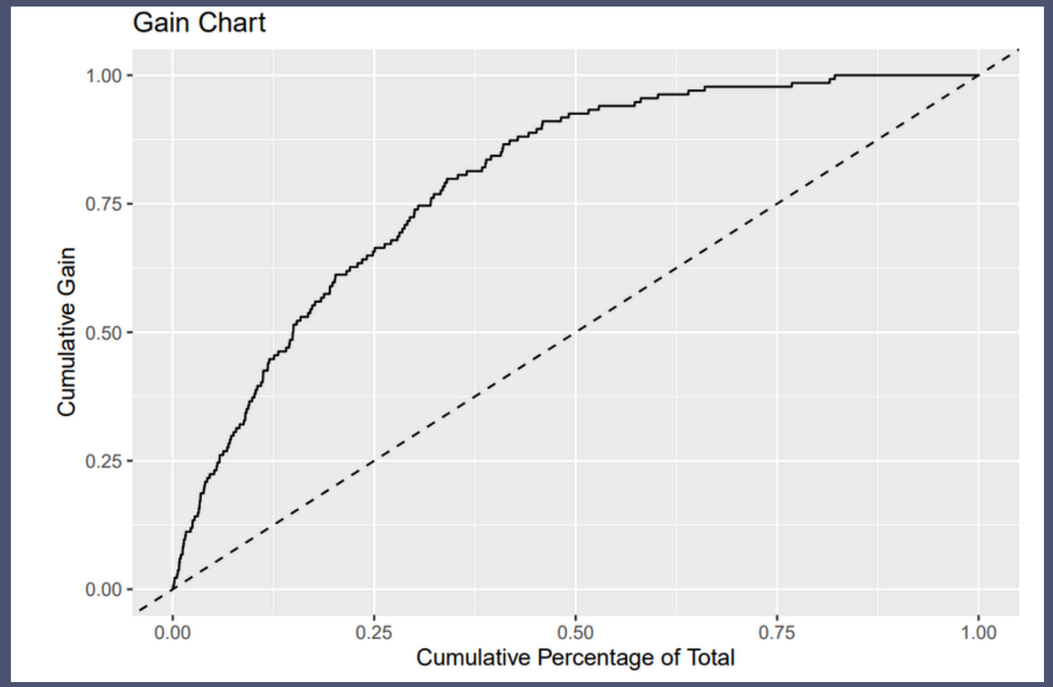
- BEST Validation F1
- BEST Validation AUC

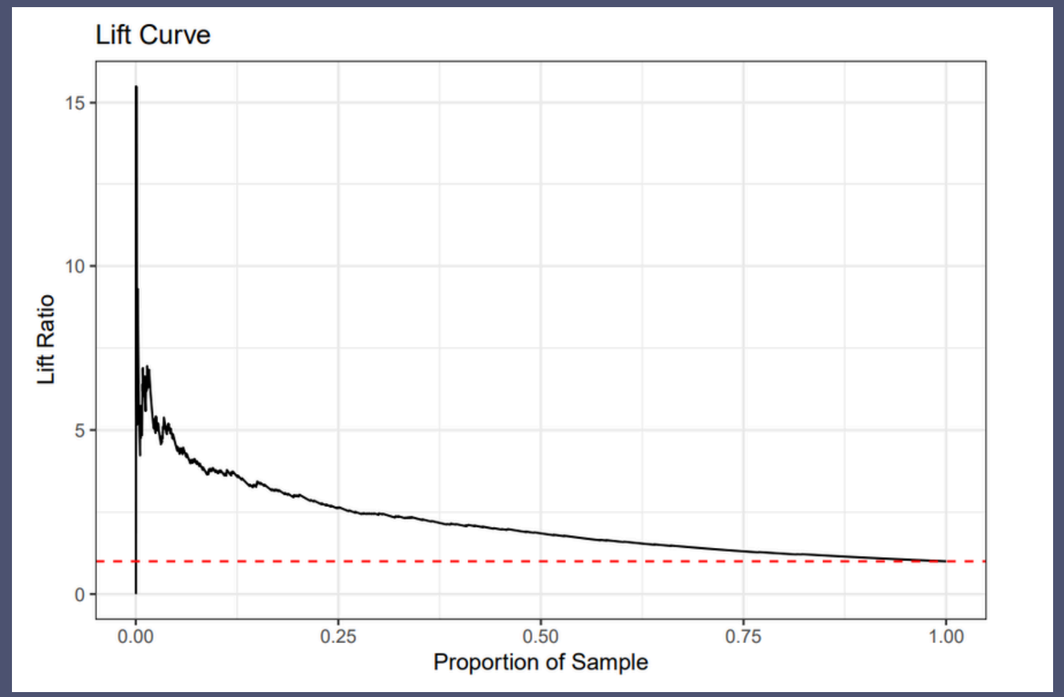
KNN	Decision Tree	Random Forest	Random Forest(Ranger)
F1: 0.08	F1: 0.13	F1: 0.16	F1: 0.21
AUC: 0.57	AUC: 0.74	AUC: 0.78	AUC: 0.82

Random Forest(Ranger) performance

Maintaining performance as in validation

- F1 = 0.18, AUC = 0.80: Stable generalization
- Gain curve/ Lift curve: Top rankers (25%) mostly captured





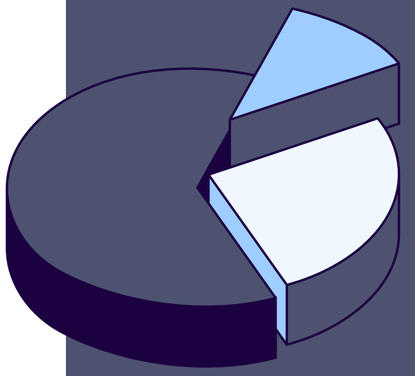
Benefits Offered:

- Best model possible for the data
- Allows you to target less people but still capture a large chunk of likely adopters - reducing marketing campaign cost
- Improve ROI on campaigns
- Insight on diversity of adopters

Evidence:

- Scenario:
  - Marketing cost per targeted user: $1
  - Subscription value per adopter: $50
    - Our model: targets 304 most likely adopters - **ROI: 12.65**
    - No model: targets all 9232 users - **ROI: 1.67**

- There is consistently higher variation (measured by sd) in the adopter class compared to non-adopters

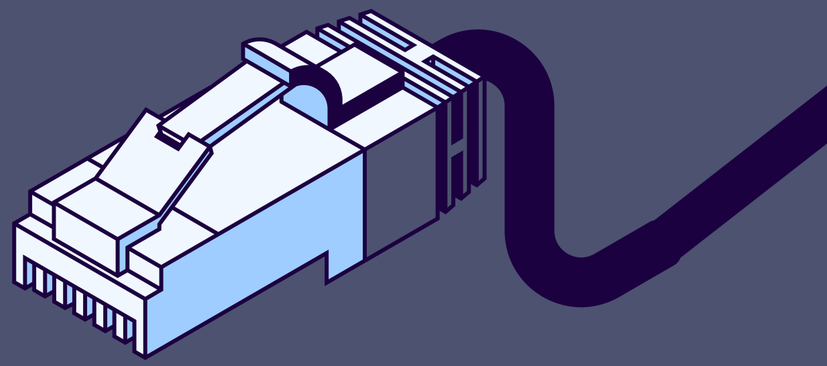


Overall:

Utilizing our model will be good for the company because it has the potential to increase ROI for the next marketing campaign.

Looking Forward:

- More data collection will only make the model better
  - Are there similar characteristics of adopters that we are missing?



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Other Recommendations:

- Incentivizing adopters in invite friends to use the platform
- Focusing R&D on song/playlist recommendations

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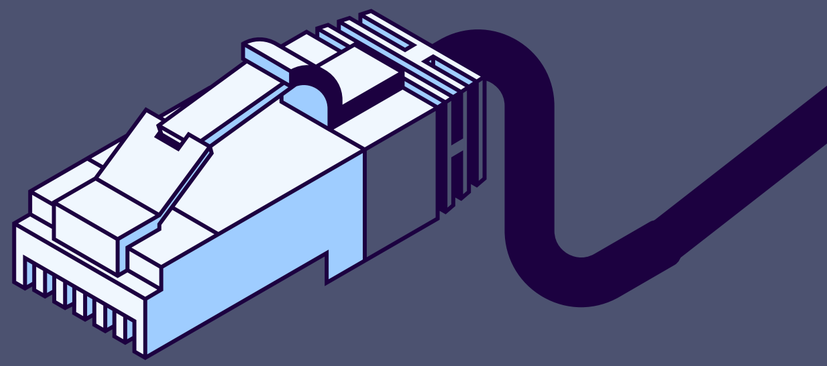

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